

Indonesia Stock Market Paper Auto Trading System Guide

What do we need to make paper auto trade system that follow big money flow in Indonesia stock market

To build a paper (simulated) automated trading system that follows "big money flow" in the Indonesia Stock Exchange (IDX/BEI), you need data access, strategy logic, a backtesting/paper trading engine, and reliable infrastructure. "Big money flow" typically refers to tracking institutional/foreign investor activity (e.g., net foreign inflows/outflows), dominant broker accumulation/distribution (known locally as Bandarmology), large block trades, and related volume/ownership shifts.

This is feasible for retail developers using public/third-party APIs, but real-time precision and full order execution simulation have limitations compared to US markets. Here's a breakdown of what you need.

1. Market Data Sources (Critical for Big Money Signals) You need reliable real-time/historical prices, broker summaries, foreign flow, and volume data.

- Invezgo API (highly recommended for Indonesia-specific features): Real-time OHLCV, broker analysis (Bandarmology), foreign flow, top foreign accumulation/distribution, KSEI shareholder data, technical indicators, and financial reports. SDKs available (Python, Go, etc.). Strong on broker summary and foreign investor tracking. - Sectors.app API: Good for foreign flow by symbol, ownership movements, and broker data. - Official IDX Data Services: Real-time/delayed/EOD data via licensed redistributors (e.g., Bloomberg, Refinitiv). More expensive/enterprise-oriented. - Other options: RapidAPI IDX wrappers, OHLC.dev, iTick, or public IDX pages (broker summary, trading summaries) scraped/parsed (less reliable for automation).

Big money proxies in IDX: - Foreign Flow/Net Foreign Buy-Sell: Daily/ cumulative tracking via KSEI or broker data. - Broker Summary/Bandarmology: Identify dominant brokers accumulating (buying) or distributing (selling) specific stocks. Tools like Stockbit or APIs show this. - Volume spikes, block trades, ownership changes. - Combine with fundamentals (financial reports) and technicals (RSI, MACD, moving averages).

Free/delayed data works for initial testing; paid APIs (~real-time) are better for live paper trading.

2. Strategy Logic: "Follow the Big Money" Define rules the system follows automatically: - Buy when foreign/broker net buying exceeds threshold + positive price/volume confirmation (e.g., accumulation zones). - Sell on distribution signals, trailing stops, or risk rules. - Filters: Liquidity (top

stocks like LQ45/IDX30), sector rotation, avoid low-volume traps. - Risk management: Position sizing, stop-loss, max drawdown. - Optional: Machine learning for pattern recognition in flows, or simple rules-based (e.g., SMA crossover + foreign inflow filter).

Start simple — backtest thoroughly before automating.

3. Tech Stack for Automation & Paper Trading - Programming Language: Python (easiest with pandas, NumPy, TA-Lib, Backtrader/Backtesting.py, or Zipline for backtesting). Use SDKs for data. - Data Pipeline: Fetch data via API (REST/WebSocket), store in SQLite/PostgreSQL for historical analysis, or use Redis for real-time caching. - Backtesting Engine: Test strategy on historical data (e.g., 5–15+ years from APIs). Evaluate Sharpe ratio, win rate, drawdown. - Paper Trading/Simulation: - IDX Virtual Investing / IDX Mobile simulator for manual practice (real market feel, virtual money). - Custom simulation in code: Track virtual portfolio, simulate fills based on real quotes (no native easy paper API like Alpaca for IDX). - Broker APIs (e.g., some local securities firms) may offer simulation accounts; integrate if available. - For algo execution logic, build an order management system (OMS) that "places" simulated orders. - Scheduling/Orchestration: Cron jobs, Airflow, or cloud functions (AWS, GCP, or local server) to run periodically (e.g., end-of-day or

intraday). - Visualization/Dashboard: Streamlit, Dash, or Grafana for monitoring signals, portfolio performance, and flow charts.

4. Infrastructure & Operations - Hardware/Hosting: Reliable server/VPS (low latency to IDX), stable internet. Cloud for scalability. -

Regulatory/Compliance: Paper trading avoids most issues, but study OJK/IDX rules if moving to live. No real orders = lower barriers. - **Costs:** API subscriptions (Invezgo/Sectors vary; start cheap), data fees, cloud hosting. Backtesting is mostly free. - **Security:** API keys, error handling, logging. Avoid over-optimization (curve-fitting).

5. Development Steps

- 1. Research & Data Access:**

Sign up for Invezgo or similar, pull sample foreign flow/broker data for key stocks (e.g., BBCA, TLKM).

- 2. Define & Backtest Strategy:** Code rules, test on historical data.
- 3. Build Simulation Engine:** Virtual

account balance, order simulation with realistic slippage/spreads.

- 4. Automate:** Fetch data →

Generate signals → Simulate trades → Log/monitor performance.

- 5. Iterate:** Paper trade in real-time,

refine based on results. Add alerts (Telegram/Email).

- 6. Risks & Realism:** Simulate commissions, taxes, liquidity, and market impact. Big money signals can lag or reverse.

Challenges Specific to Indonesia - Foreign flow data is strong but sometimes delayed or aggregated. - Less mature algo ecosystem than US (no seamless Alpaca-like paper API). - Market specifics: High retail participation, foreign dominance in some stocks, trading hours (Jakarta time), T+2 settlement. - Data quality: Use official/verified sources to avoid garbage-in-garbage-out.

Getting Started Recommendation: Use Python + Invezgo SDK for data, Backtrader for backtesting, and a custom simulator. Prototype on a few liquid stocks. Communities like Stockbit or local quant forums can help with Bandarmology nuances.

This setup can evolve from paper to live (via a broker API/integration) once proven. Success depends on robust strategy, discipline, and continuous validation — no system guarantees profits. Start small and paper trade extensively.